

FM3 Requirement Specifications: Edge Algorithm

Combined Foreign Matter Detection (FMD) and Missing/Multiple (MMLS)
Vision Systems

Revision Level: Draft

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Section 1: Introduction

1.1 Current State

The production environment includes around one hundred automated manufacturing lines operating 24 hours a day, seven days a week, producing over 35,000 unique SKUs. Each of these manufacturing lines has a dedicated Foreign Matter Detection (FMD) Vision System and a separate Missing/ Multiple Lens Detection System (MMLS).

1.1.1 FMD

The goal of FMD is to discard any packages found to contain Foreign Matter (FM) which is defined by the quality procedure QP-0050, as non-lens material in contact with the packing solution which is greater than or equal to 647 microns in the largest dimension. There are currently 5 major FMDVS families of variation at the Deerwood Jacksonville Site, where the setup, number of cameras and illumination changes between each, with each line in 2GT running 5 cameras, 3GT Legacy lines running 6 cameras (pre-dosing), 3GT Flex lines running 1 camera, 4GT lines running 10 cameras, and the 5GT line running 10 cameras. All these inspection systems take the image after dosing (except for the 3GT legacy lines which take the image right before dosing) where they perform the classification analysis and then push raw image data and analysis to the line server.

The system is also capable, but not currently validated, of detecting and rejecting packages with missing lenses, multiple lenses, and some lens defects. The current FMDVS is only capable of identifying whether FM or debris exists within the package at an 80% capability requirement. This necessitates a secondary, manual visual inspection via sampling in the Quality Control (QC) lab. This also means that the system cannot return any data regarding what type of FM or defect was detected.

The FMD system analyzes digital images captured by a grayscale camera while the contact lens is in an unsealed primary package with the final packaging solution and in motion on the automated manufacturing line. The manufacturing machine then rejects the array of lenses after it is heat sealed if the FMDVS determines it contains FM/debris.

1.1.2 MMLS

The goal of MMLS is to discard packages that do not have a contain a lens or have multiple lenses. There are currently 5 major MMLS families of variation at the Deerwood Jacksonville Site, where the setup has slight changes between each. The big differences are when the packages are inspected and how the defects are sorted. On the manufacturing lines in 2GT, 4GT and 5GT, the MMLS inspection occurs from above on unsealed, dry packages right before dosing while in all the 3GT lines the inspection occurs from below on the sealed packages right before sterilization. From a sorting perspective, the manufacturing lines in 4GT, 5GT and on 2GT lines 3 and 4 the defects are sorted and rejected individually while on

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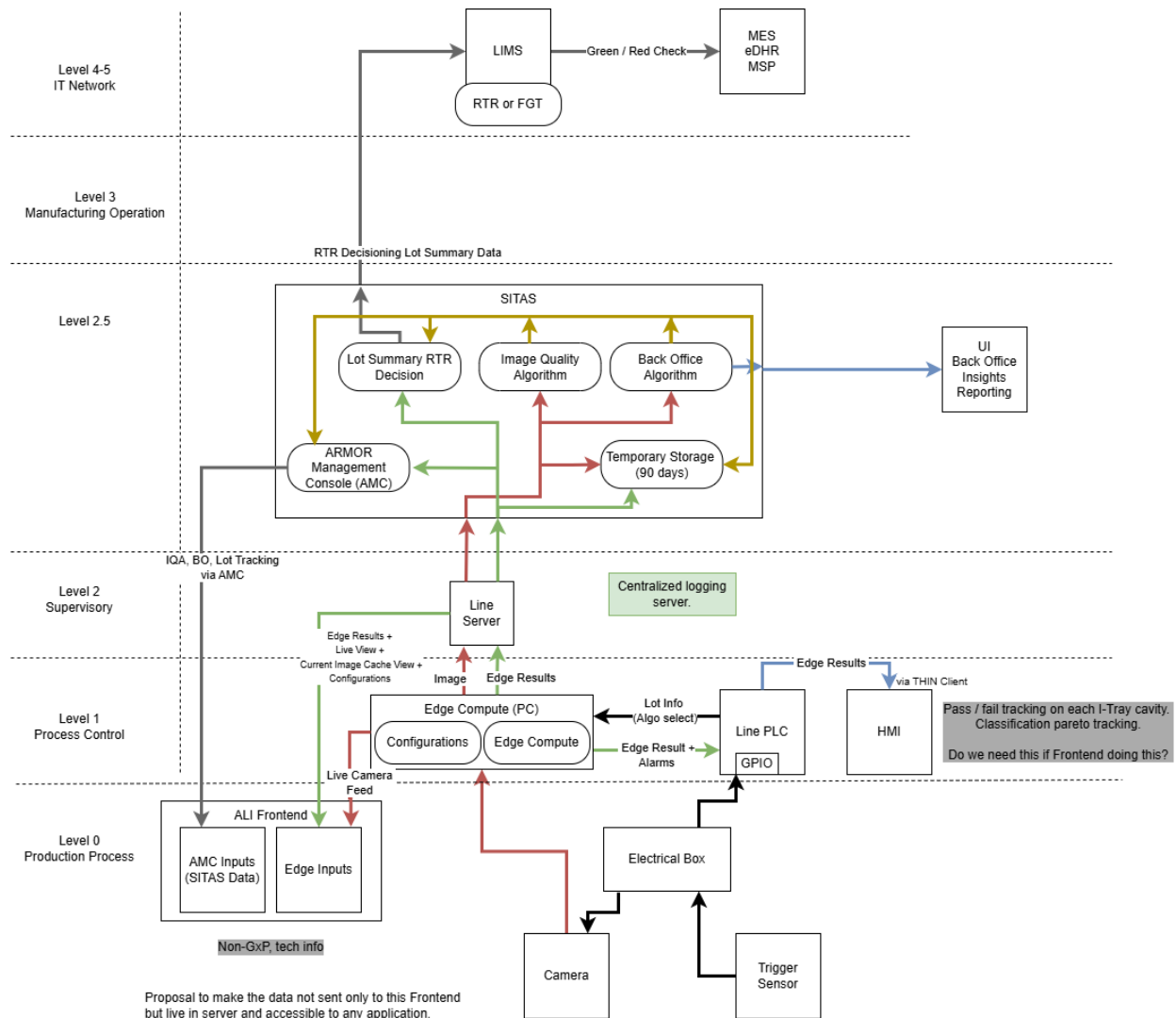
the rest of the 2GT lines and in 3GT the defects are rejected per array group, so one defect can throw away up to 30 sealed packages depending on the setup.

1.2 Purpose

This document provides the business and technical functional requirements for the planned combined FMD/MMLS algorithm evaluation referred to as the FM3. The scope of FM3 will include an additional function compared to the FMD, of being able to differentiate between FM/other defects and will have full capability of detecting packages with missing/multiple lenses. The current FMD image formation will be utilized to complete this evaluation and will not change for rollout of FM3.

The purpose of the FMD algorithm evaluation is to identify the algorithm and verify that the current hardware is capable of meeting production requirements including correct classification and cycle time. Furthermore, identify key learnings that will have a high probability of being transferrable to the new image solution. For example, which types of debris are most difficult for the system to classify and why. Image formation, manufacturing integration, data management, support infrastructure, and operator interfaces are out of scope for the FMD algorithm evaluation.

1.3 Future State Infrastructure



Section 2: Locations

Johnson & Johnson MedTech facilities for manufacturing contact lenses are in Jacksonville, Florida, and Limerick, Ireland.

Section 3: References

Document Number	Description	Current Revision
QP-0050	Quality Assurance Program for Products Using and Alphanumeric Lens Lot Number at the Deerwood Park Facility	115
QP-0205	Global Quality Assurance Program	22
VWA-0523	Actions Required for Product Involved in Quality Events	44
VWA-0795	Cartons, Foil and Primary Package Defect Classification & Bounding Reference Document	52
VWA-1155	Product Quality Attribute (PQA) List	44
02IG-4029	2GT Foreign Matter Detection (FMD) Technical and Maintenance Manual	3
03IG-2007	3GT FMD System TAM1-23 and Tam Flex Lines	1
04IG-4022	4GT Foreign Matter Detection Technical and Maintenance Manual	1
05IG-4020	5GT Foreign Matter Detection Technical and Maintenance Manual	1
IOIDP-0585	2GT EzEye FMD Technical & Maintenance Manual	4
IOIDP-0591	3GT Integro Foreign Matter Detection Vision System Manual	3
IOIDP-0600	4GT Hybrid/5GT EzEye Foreign Matter Detection Technical and Maintenance Manual	3
VAL-0009	Validation of Lens Quality for Visual Attributes	14
VAL-0028	Qualification of the Foreign Matter Detection Vision System	1

Section 4: User Requirement Specifications

This document has two tiers of URS items which are all driven by our quality documentation. The first is requirements which are criteria this system must meet to be considered successful. The second is nice-to-haves which are not required by our quality documentation but will increase the detection capability and will help the manufacturing lines produce a more pristine product. These nice-to-haves will also help to decrease the number of complaints and non-conformances, so wherever possible, they should be implemented.

4.1 Requirements:

1. Detection of objects on the lens or in solution that are 647um or larger
2. Detection of missing or multiple lens defects
3. Differentiating between lens material defects (hema rings, excess pieces, extraneous polymer, etc.) from other defects
4. Differentiation of bubbles and fibers from other defects
5. Record of the time of inspection

4.2 Nice to Haves:

1. Detection of objects on the lens or in solution that are smaller than 647um and grouping of these defects by size range:
 - a. < 300um
 - b. 300 - 500um
 - c. 500 - 647um
2. Differentiation between other lens visual defects (lens edge defects, surface marks, etc.) from other defects
3. Material analysis to determine the composition of an object in question to better help classify, e.g. defect morphology

Section 5: FMD Images

The FMD system captures visible light images of each lens under illumination from a strobe below the carrier.

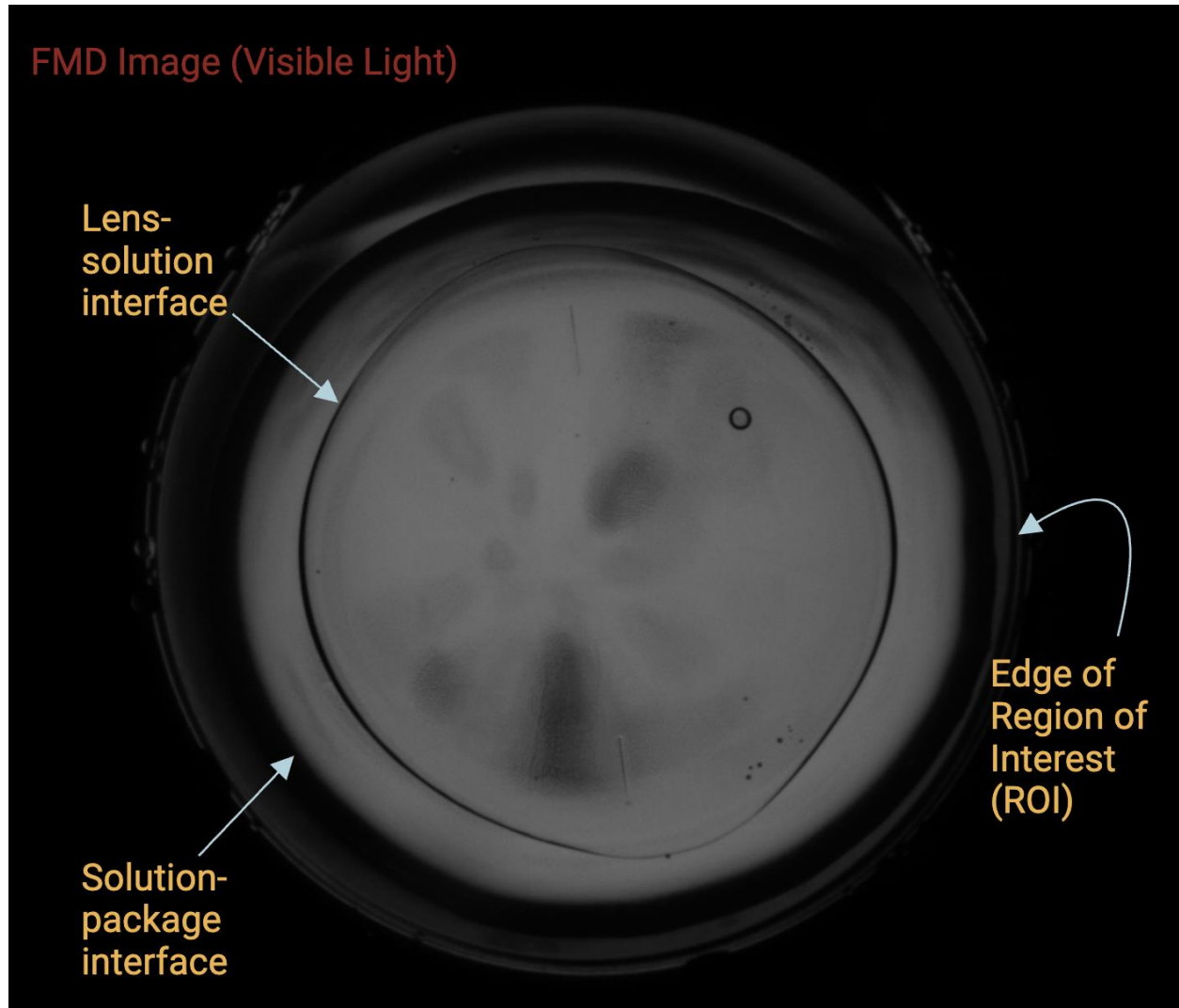


Figure 1. Sample inspection image showing key components and regions of interest.

5.1 Bowl Presentation

The current FMD system does not require a pristine bowl presentation, but FM3 will require a near-pristine bowl presentation. Multiple factors are involved in determining if bowl presentation is adequate for product inspection, and currently this determination is subjective. In scope of this algorithm the input parameters to determine if the bowl presentation is adequate to identify debris or other artifacts include:

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- Wet package
- Dirty strobe
- Dirty camera
- Bubble scatter
- Low dose obstructing the region of interest
- Missing Primary Package
- Primary Package Misalignment
- Field of View Obstruction
- Lens Off Center
- Hema Obstruction
- Missing Lens
- Multiple Lens

Additional bowl presentation defects that will be handled internally include:

- Camera Focus
- Image Brightness
- Primary package marking
- Ripple obstruction
- Bubble obstruction
- Incomplete image capture

5.1.1 Wet Package

This form of bowl presentation issue presents itself when the underside of the primary package becomes wet. It appears as irregular bubbles but is really wetness under the package (Figure 2).

Key Characteristics of wet package:

- Reduced clarity in affected regions
- Localized impact on image quality / light distribution

Impact on inspection quality:

- Compromised debris detection accuracy
- Increased false positive rates in automated detection
- Masking of genuine defects
- Reduced reliability of automated measurement

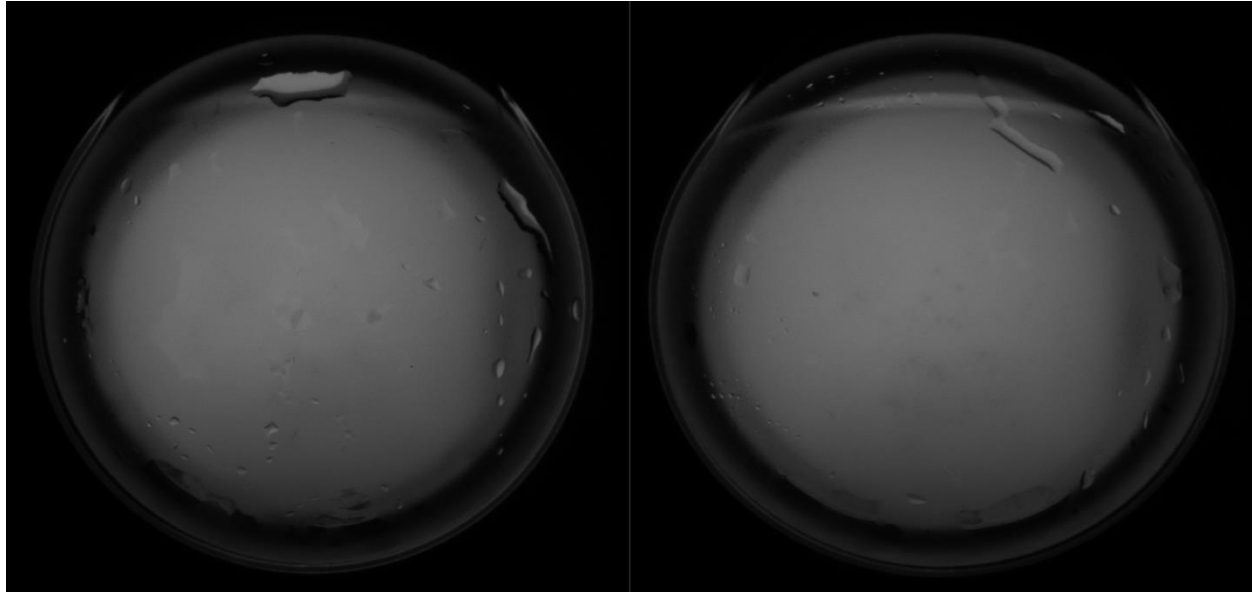


Figure 2. Sample inspection images showing examples of wet primary packages.

5.1.2 Dirty Strobe

Dirty strobe occurs when saline or other substances build up on the cover between the strobe and the bottom of the package passing over it. This manifests in the FMD image as out-of-focus and dark marks (Figure 3).

Key characteristics of dirty strobe:

- Dark, out-of-focus marks that appear behind the package
- Consistent across consecutive images
- Resulting shape does not extend outside of illuminated region
- Reduced contrast in affected regions
- Localized impact on image quality

Impact on inspection quality:

- Compromised debris detection accuracy
- Masking of genuine defects
- Risk of false positive artifact detection
- Challenge for automated detection algorithms
- Need for position-independent recognition

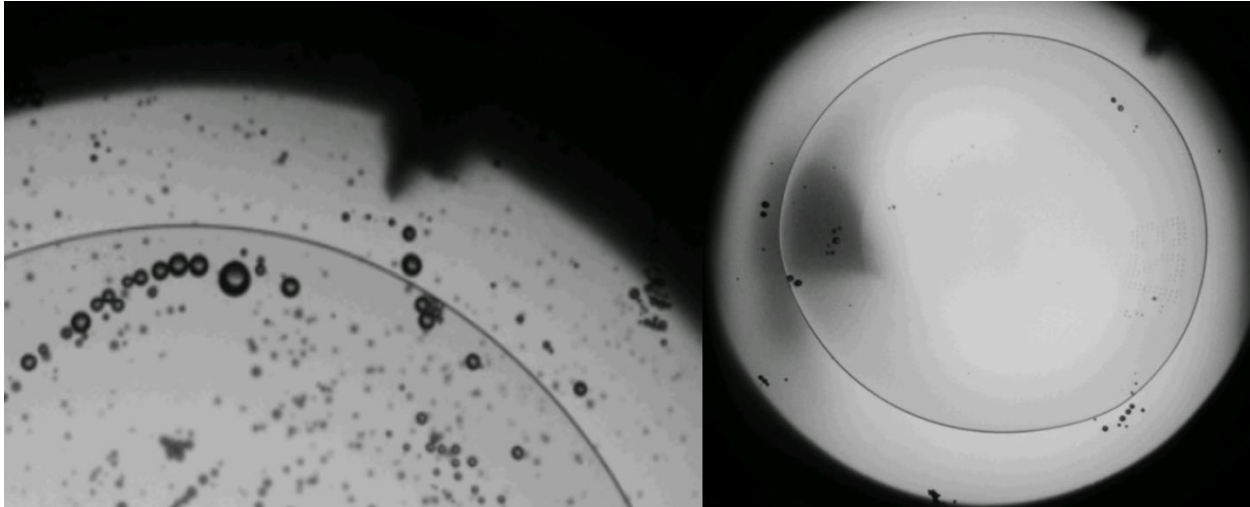


Figure 3. Sample inspection images showing different manifestations of a dirty strobe.

5.1.3 Dirty Camera

A dirty camera is most often due to dried saline on the camera lens or camera lens cover. This differs from a dirty strobe as a dirty camera results in less dark marks that can extend outside of the illuminated region of the image (Figure 4).

Key characteristics of dirty camera:

- Out-of-focus marks that appear on top of the package
- Consistent across consecutive images
- Cannot be altered by lens or package appearance
- Resulting shape may extend outside of illuminated region
- Reduced contrast in affected regions
- Localized impact on image quality

Impact on inspection quality:

- Compromised debris detection accuracy
- Masking of genuine defects
- Risk of false positive artifact detection
- Challenge for automated detection algorithms
- Need for position-independent recognition

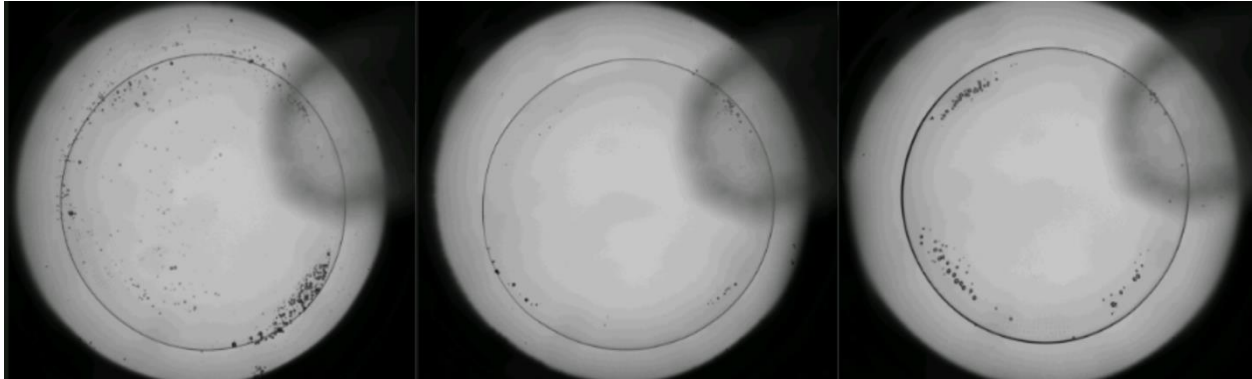


Figure 4. Sample images showing a dirty camera mark across 3 different inspection images.

5.1.4 Low Dose

Low dose is an obstruction that results from the final saline dose failing to spread across the entirety of the bowl. This causes a region or regions of dark shading that can mask debris or other artifacts (Figure 5).

Key characteristics of low dose:

- Exceedingly dark regions near lens edge
- Reduced clarity in affected regions
- Localized impact on image quality / light distribution

Impact on inspection quality:

- Compromised debris detection accuracy
- Increased false positive rates in automated detection
- Masking of genuine defects

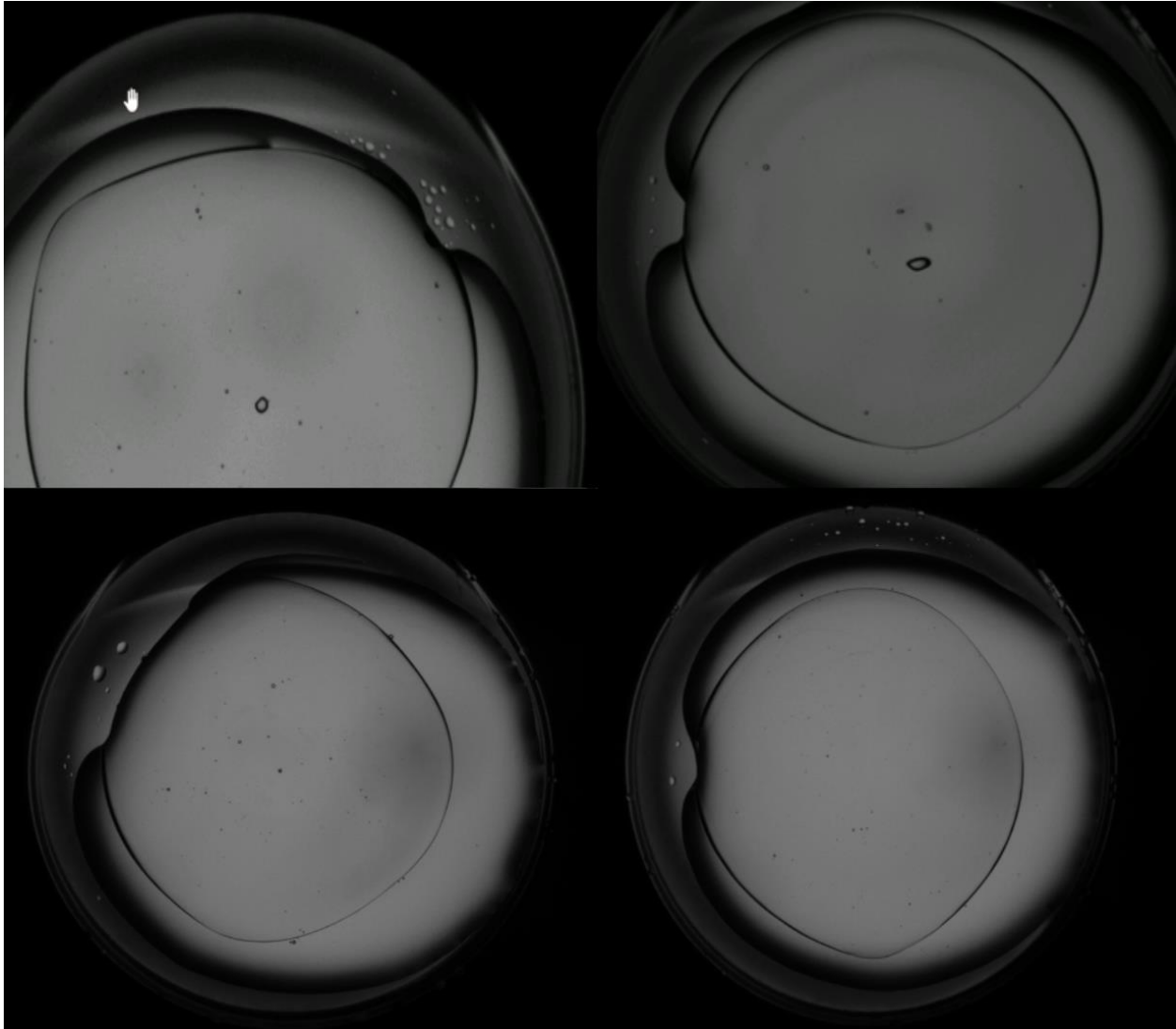


Figure 5. Samples FMD images showing low dose.

5.1.5 Bubble Scatter

Bubble scatter presents as a field of many small bubbles beneath the lens surface, creating significant optical interference that impedes accurate inspection (Figure 6). This phenomenon differs from individual bubble obstructions by creating a diffuse pattern of interference across the field of view.

Key characteristics of bubble scatter:

- Multiple small bubbles distributed throughout the bowl
- Diffuse optical interference pattern
- Reduced contrast in affected regions

- Non-localized impact on image quality

Impact on inspection quality:

- Compromised debris detection accuracy
- Increased false positive rates in automated detection
- Masking of genuine defects
- Reduced reliability of automated measurement

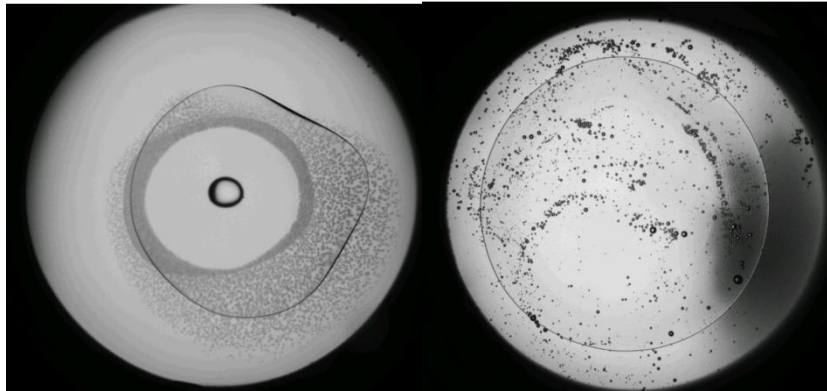


Figure 6. Sample inspection images showing bubble scatter patterns affecting lens inspection.

5.1.6 Missing Primary Package

Missing primary package defects present as a brighter image, usually verging on oversaturated, due to the absence of the primary package bowl which acts as a diffuser (Figure 7).

Key characteristics of missing primary package:

- Brighter image relative to the surrounding images
- Only defined features are the light/dirty camera marks
- Missing lens

Impact on inspection quality:

- No debris detection accuracy
- Increased false positive rates in automated detection

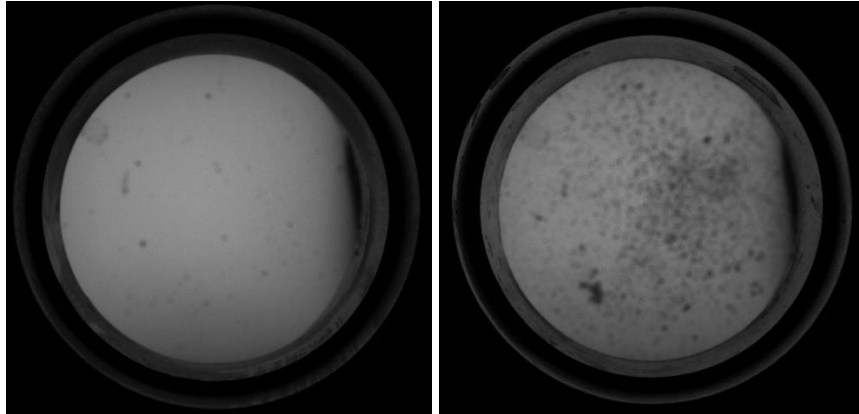


Figure 7. Sample inspection images showing missing primary packages affecting lens inspection.

5.1.7 Primary Package Misalignment

Primary package misalignment presents as off center images of the bowl, usually showing the heat seal area and edge of the package which is not usually seen. The image also does not have any full circular shapes like the light and lens which are normally seen (Figure 8).

Key characteristics of primary package misalignment:

- Normally unseen package features such as the heat seal and package edge
- Defined shadows from the package
- Reduced contrast in affected regions
- Localized impact on image quality / light distribution

Impact on inspection quality:

- Compromised debris detection accuracy
- Increased false positive rates in automated detection
- Genuine defects unable to be seen offscreen

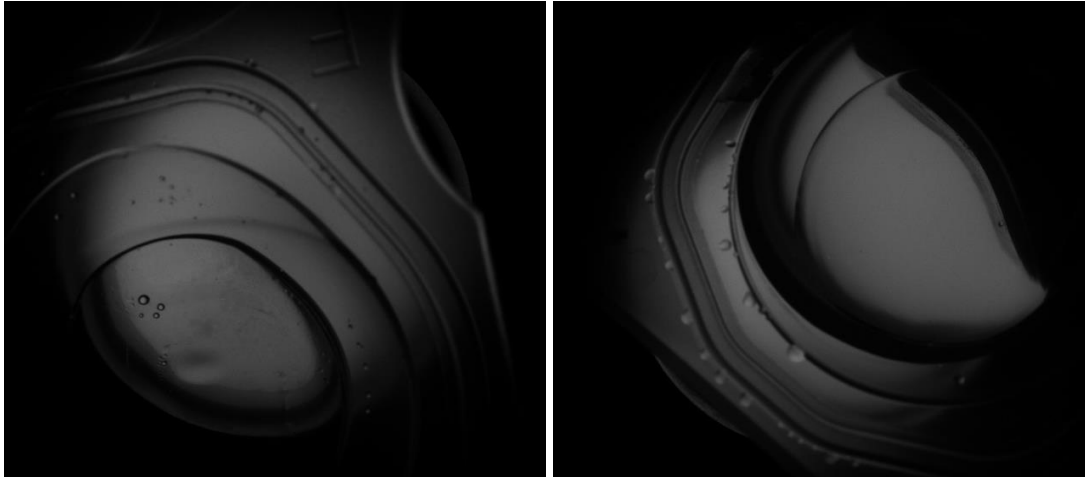


Figure 8. Sample inspection images showing misaligned primary packages affecting lens inspection.

5.1.8 Field of View Obstruction

Field of View Obstruction presents as the image being partially or in some cases fully obstructed by the foil or some other object. This causes a region or regions of near complete darkness that can mask debris or other artifacts making defect detection impossible in those darkened regions (Figure 9).

Key characteristics of primary package misalignment:

- Cut off or absence of circular lens boundary
- Dark sections of the image
- No contrast in affected regions
- Localized impact on image quality
- In most cases, defined straight edges of the obstructing object

Impact on inspection quality:

- Compromised debris detection accuracy
- Increased false positive rates in automated detection
- Genuine defects unable to be seen offscreen

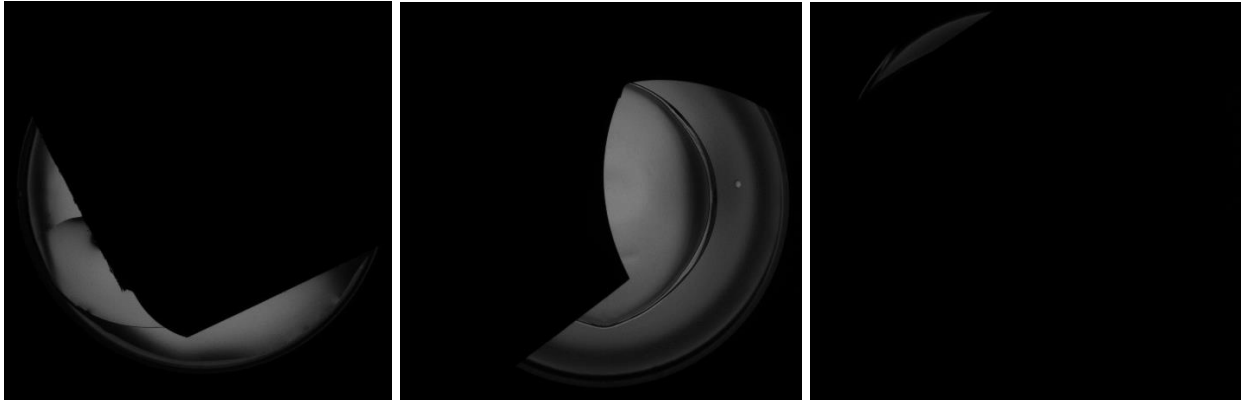


Figure 9. Sample inspection images showing varying degrees of field of view obstructions which affect lens inspection.

5.1.9 Lens Off Center

Lens off center defects appear as part of the lens edge being darkened which can mask debris or other artifacts. It appears similar to low dose defects and usually occurs when the lens bends with a convex meniscus of the packing solution on or close to the edge of the bowl (Figure 10).

Key characteristics of primary package misalignment:

- Obstructed or dark regions near lens edge
- Incomplete lens boundary
- Reduced clarity in affected regions
- Localized impact on image quality / light distribution

Impact on inspection quality:

- Compromised debris detection accuracy
- Increased false positive rates in automated detection
- Masking of genuine defects

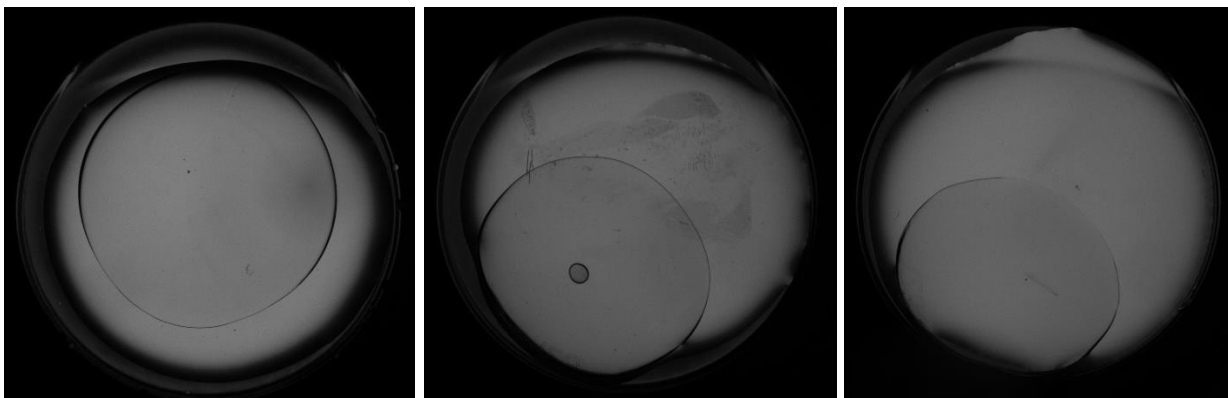


Figure 10. *Sample inspection images showing lens off center defects*

5.1.10 Hema Obstruction

See HEMA category in section 5.2.1.

5.1.11 Missing Lens

Missing lenses can occur due to several upstream issues and must be identified as a failure before heat seal (Figure 11).

Key Characteristics:

- Absence of circular lens boundary
- Uniform background appearance
- Visible package features only
- Lack of characteristic lens edge shadows
- Saline solution artifacts may be present

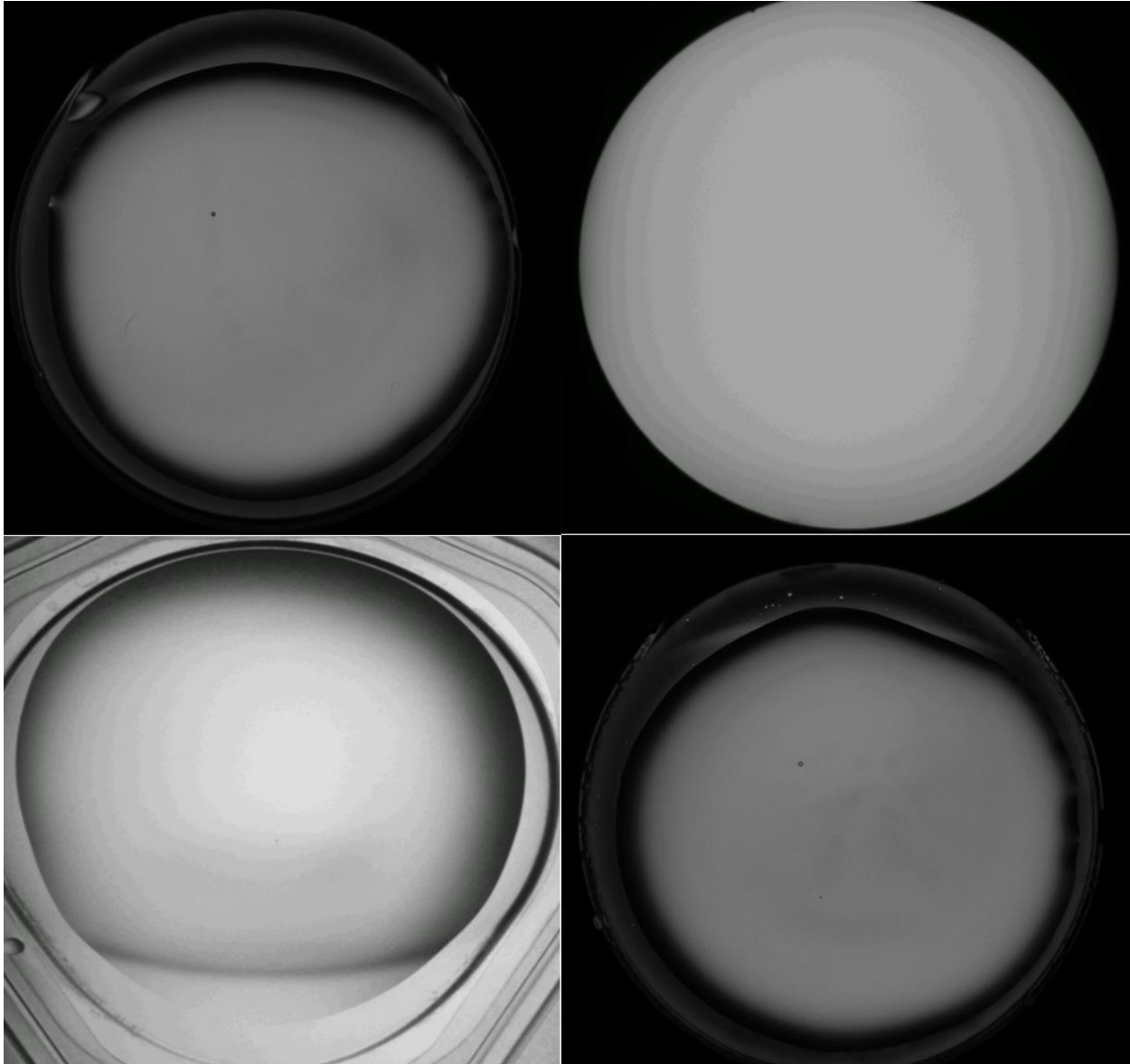


Figure 11. Four different examples of FMD images with missing lens and one where it is missing both the lens and primary package.

Missing lens is always considered deficient for Pass/Fail determination.

5.1.12 Multiple Lenses

Multiple lenses can occur due to several upstream issues and must be identified as a failure before heat seal. Multiple lenses can be as simple as two lenses in one package or may be more complex such as one lens with an additional partial lens (Figure 12) or even three or more lenses in one package.

Key Characteristics:

- Multiple distinct circular boundaries
- Overlapping or separate lens positions
- Potential for lens-to-lens interaction
- Multiple "123" or fiducial marks may be visible

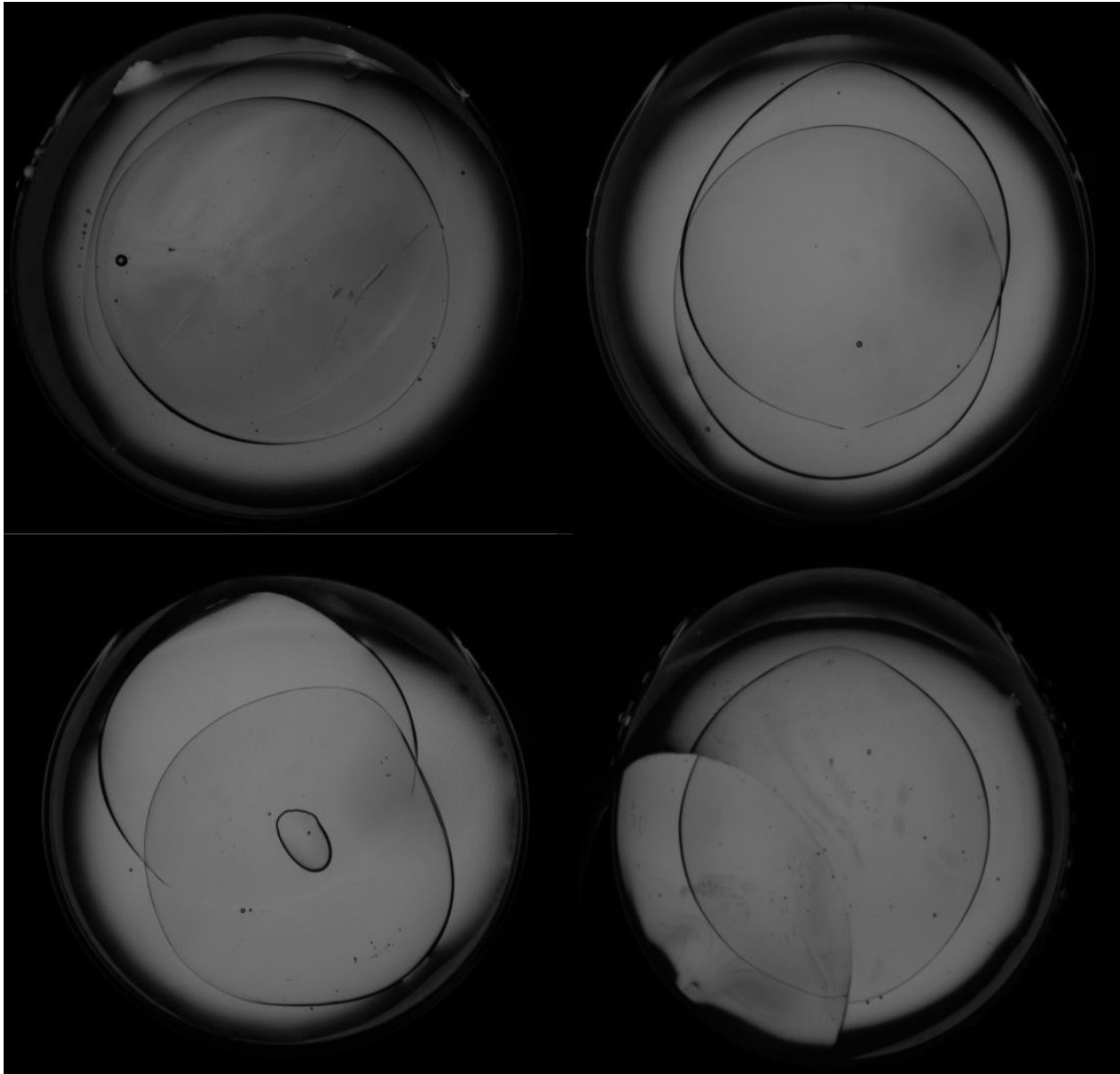


Figure 12. Four sample images showing instances of multiple lenses.

Multiple lenses are always considered deficient for Pass/Fail determination.

5.1.13 Primary Package Mark

Primary package marks (PPMs) are distinctive features present on the lens container rather than the lens itself. PPMs are characterized by their presence across multiple images

within a manufacturing batch, though their relative position to the lens may vary (Figure 13).

This class of presentation defects will be handled internally by our onsite image quality tools as the best way to differentiate these with other defects is via trend analysis.

Key characteristics of PPM presentation:

- Consistent presence across multiple images from the same batch
- Variable appearance due to lighting and focus conditions
- Dynamic positioning relative to lens location
- Potential resemblance to bubbles, tears, or other defect

Impact on inspection quality:

- Risk of false positive artifact detection
- Requirement for batch-level analysis to distinguish from true rejects
- Challenge for automated detection algorithms
- Need for position-independent recognition

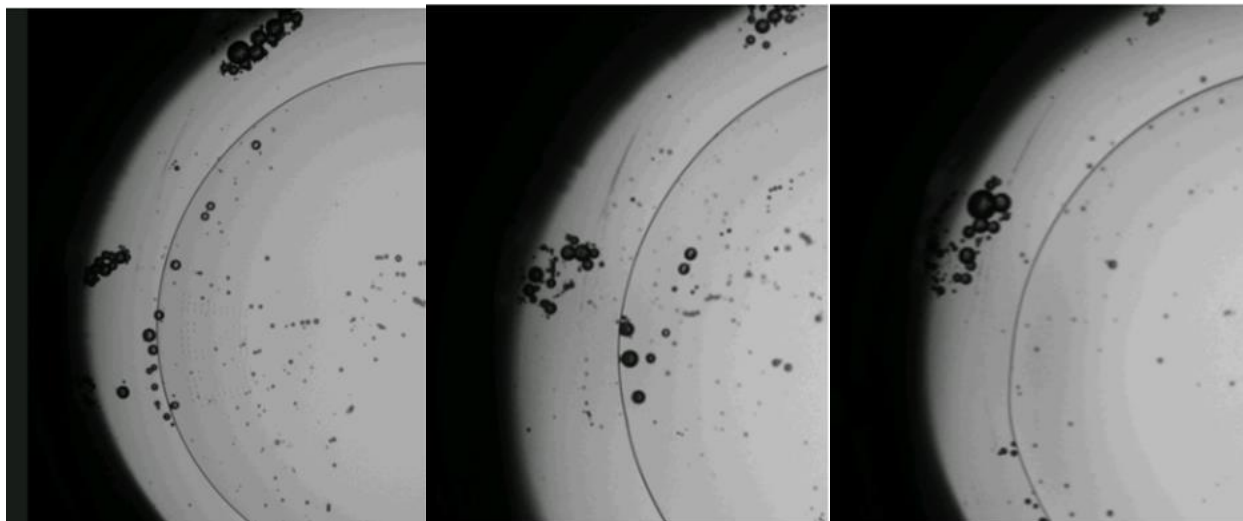


Figure 13. Set of three different sample images all containing the same primary package mark.

Technical significance:

The current FMD system's inability to distinguish PPMs from genuine debris represents a critical area for algorithmic improvement. Accurate PPM identification and classification could significantly reduce false rejects and improve overall inspection reliability.

Bowl presentation is considered adequate for Pass/Fail determination with any amount of primary package marks.

5.2 Foreign Matter and Debris

To ensure product quality and safety, foreign matter and debris require systematic detection and classification according to standardized criteria.

Detection Methodology and Classification Framework:

- Primary classification by type:
 - HEMA
 - Fiber
 - Extraneous Polymer
 - FM

Debris and foreign matter can be measured in two ways, diameter or length. Diameter is a measure of distance between the furthest two points on the debris. Length is the measure of distance along the debris from one end to the other. The diameter measurement is used for debris that lacks a clear start and end to its structure or for debris that has looped or curled back onto itself, making the length measurement too complex. The length measurement is used for debris that has a clear start and end to its structure, such as a fiber that has not curled back on itself.

5.2.1 HEMA

HEMA is the excess lens material that is removed from the lens during the demold process but can sometimes make an appearance further downstream. It can appear under the FMD camera as an entire ring or just a segment of the ring (Figure 14). A length measurement is typically used for this type of debris.

Key Characteristics:

- Ribbed pattern
- Same material as lens (translucent)
- Arc shaped
- Almost exclusively gross in size
- Sharp, well-defined edges

A subcategory of HEMA defects are those that overlap with more than 20% (pending EDA) of the visible lens will be classified under presentation defects as HEMA Obstruction as they will impact the systems ability to identify other defects under and around the HEMA.

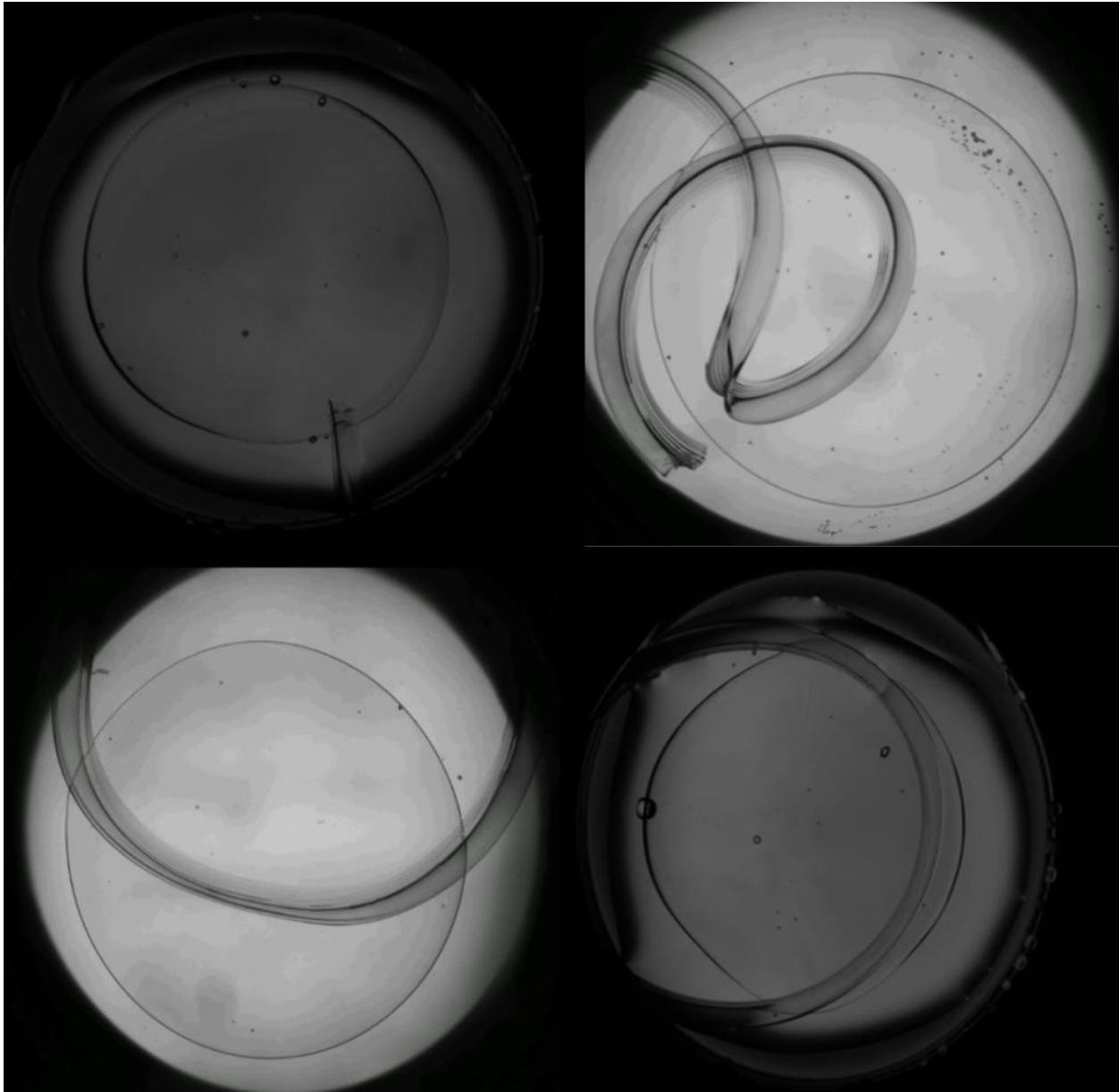


Figure 14. Sample FMD images showing full and partial HEMA rings.

5.2.2 Extraneous Polymer

Extraneous polymer is excess lens material that may appear thick or thin but is always translucent (Figure 15). A length or diameter measurement can be used for this type of debris.

Key Characteristics:

- Smooth appearance
- Same material as lens (translucent)
- Typically curling in shape
- Sharp, well-defined edges

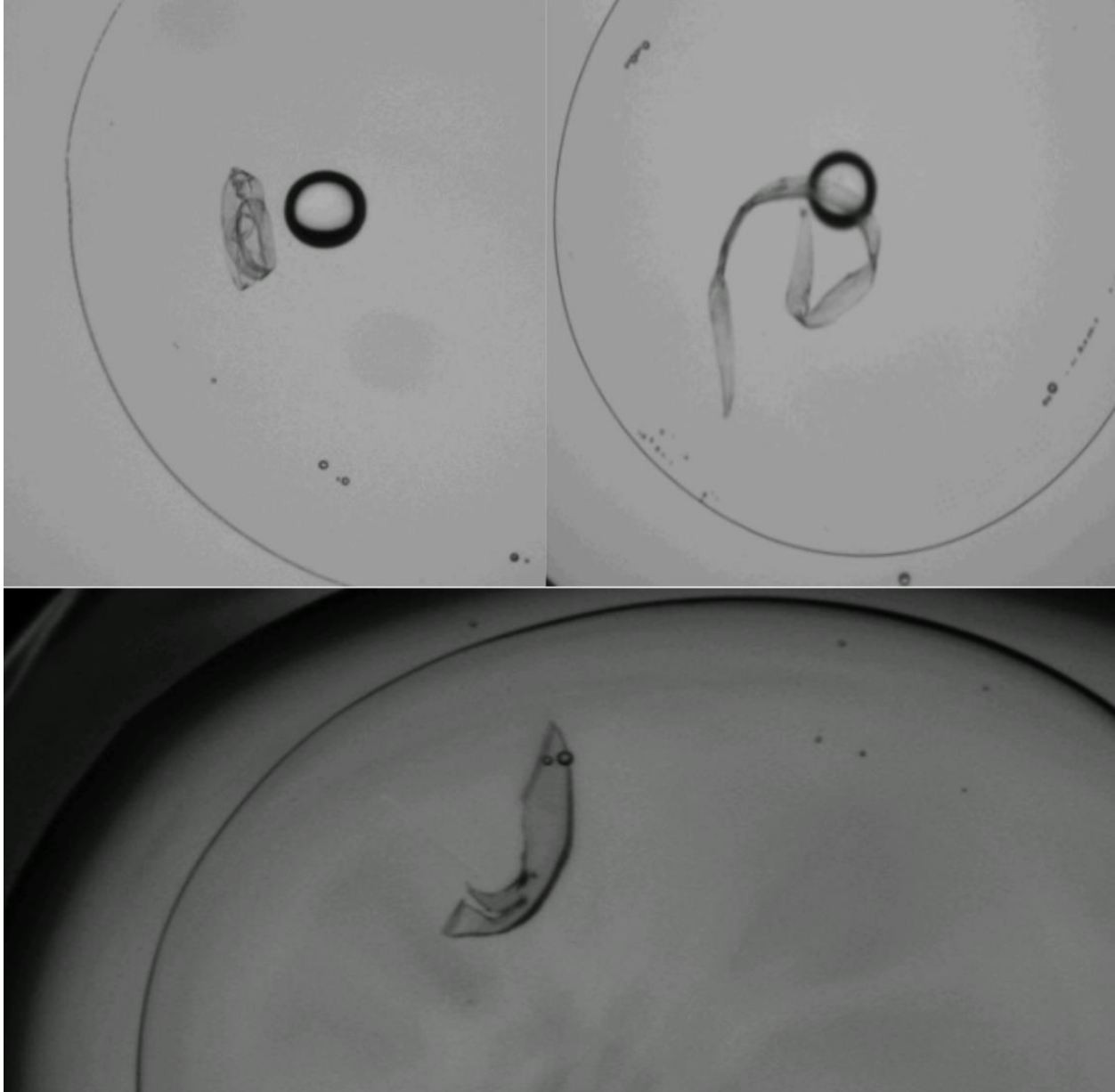


Figure 15. Three sample images showing extraneous polymer under the FMD camera.

5.2.3 Fiber

Fibers are debris that can range in size and shape but are always thin and dark (Figure 16). A length measurement is typically used for this type of debris.

Key Characteristics:

- Varying material make-up
- Thin and dark
- Varying in definition (due to depth of bowl)
- Shape varies from straight to jagged and curly

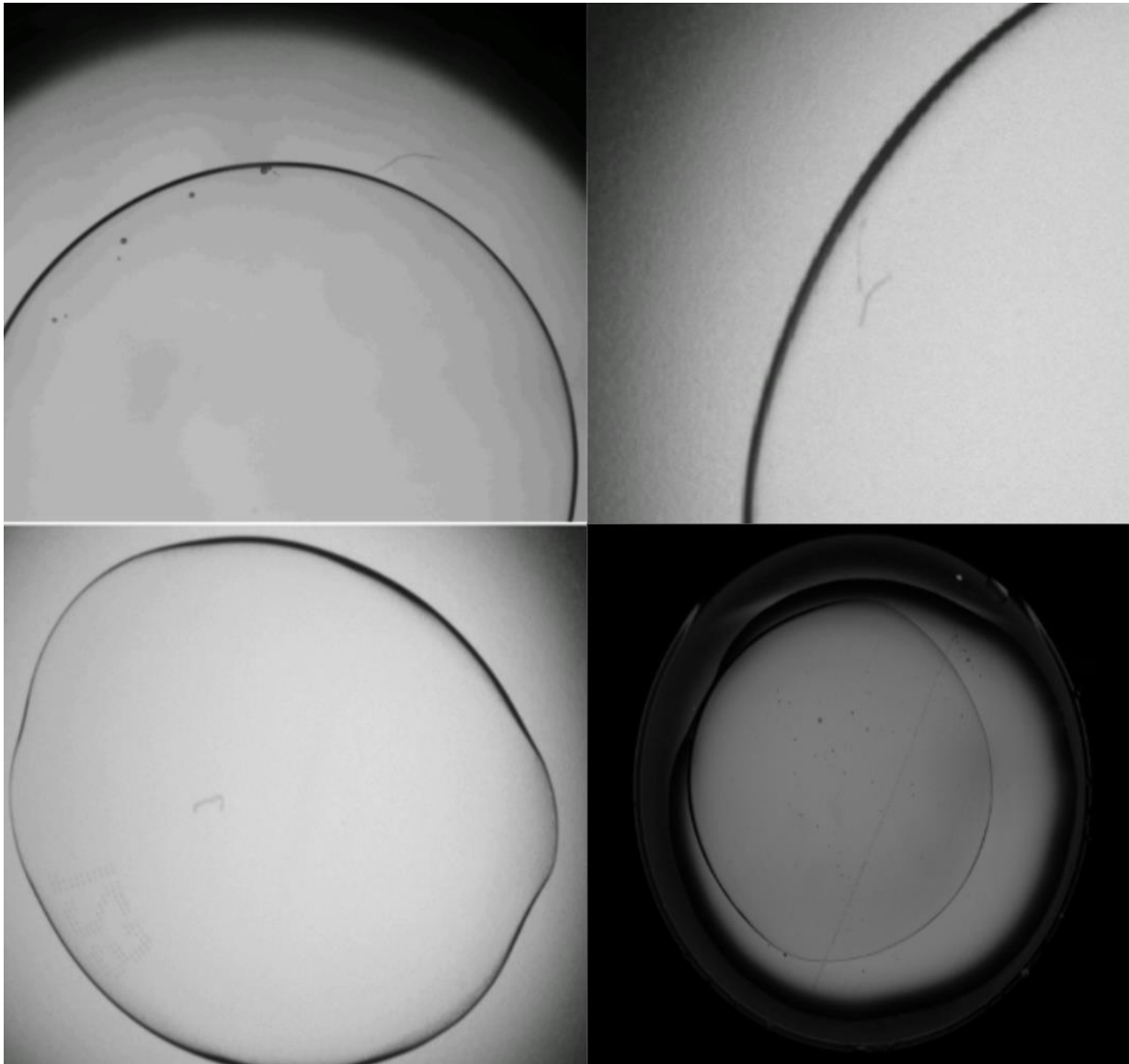


Figure 16. Four sample FMD images showing examples of fibers.

5.2.4 FM

The foreign matter (FM) category encompasses any type of debris that cannot be classified as HEMA, Extraneous Polymer, or Fiber. It can exist anywhere within the illuminated bowl and can range in size, shape, and color (Figure 17).

Key Characteristics:

- Dark, irregularly shaped objects
- Mostly opaque in nature
- Sharp, well-defined edges (distinct from bubble boundaries)
- Variable opacity levels
- Non-circular geometry

Classification:

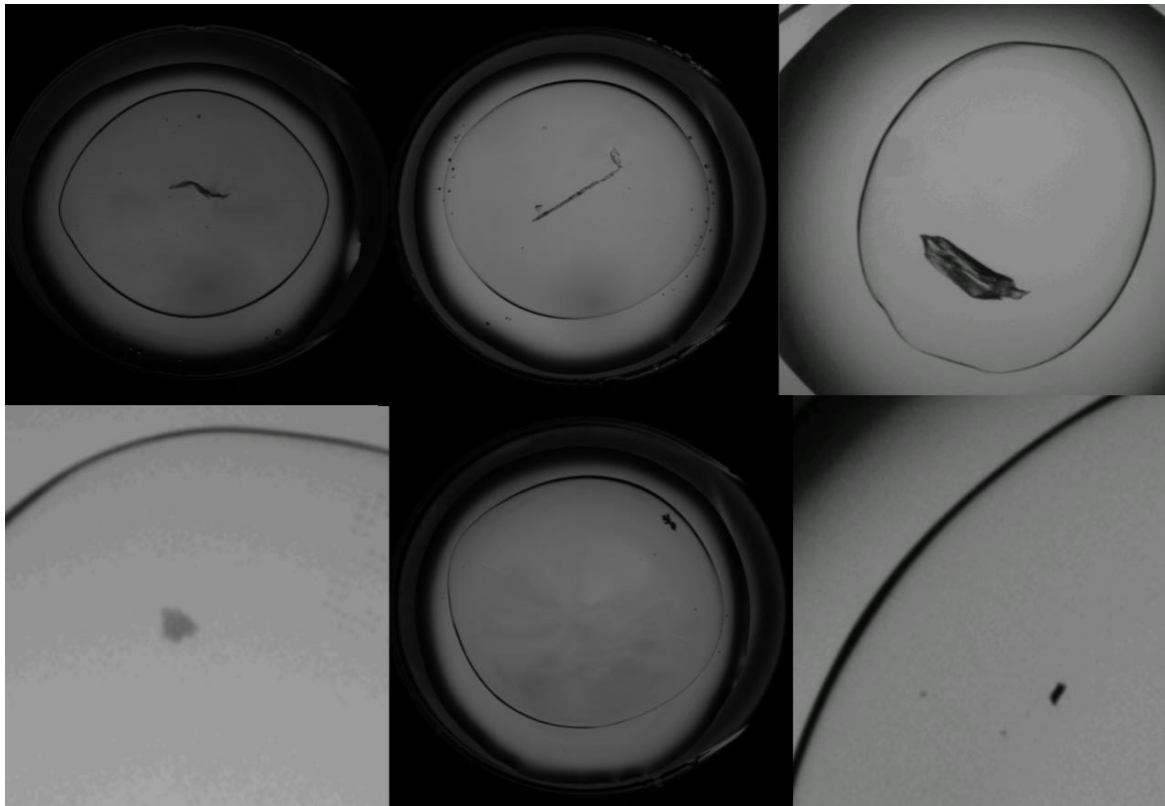


Figure 17. Sample FMD images showing different types of FM.

Section 5: Performance Metrics

The algorithm's performance in identifying the probability of pristine bowl presentation will be evaluated using the accuracy calculation.

Accuracy is defined as the following equation:

$$Accuracy = \frac{True\ Accept + True\ Reject}{Total\ Sample\ Size}$$

$$Total\ Sample\ Size = True\ Accept + False\ Accept + True\ Reject + False\ Reject$$

The algorithm's performance in identifying debris will be evaluated using the True Accept Rate (TAR) and True Reject Rate (TRR) calculations.

True Accept Rate (TAR) is the proportion of good lenses that are correctly passed by the system and is defined by the following equation:

$$True\ Accept\ Rate\ (TAR) = \frac{True\ Accept}{All\ Good\ Images}$$

True Reject Rate (TRR) is the proportion of defects correctly caught by the system and is defined by the following equation:

$$True\ Reject\ Rate\ (TRR) = \frac{True\ Reject}{True\ Reject + False\ Accept}$$

False Reject Rate (FRR) is the proportion of good lenses that are falsely rejected by the system and is defined by the following equation:

$$False\ Reject\ Rate\ (FRR) = \frac{False\ Reject}{True\ Reject + False\ Accept}$$

False Accept Rate (FAR) is the proportion of defects missed by the system and is defined by the following equation:

$$False\ Accept\ Rate\ (FAR) = \frac{False\ Accept}{True\ Reject + False\ Accept}$$

Relationship: $TRR + FAR = 100\%$ (For defect detection on defective samples)

Note: False Accept Rate can only be calculated when the test population is 100% defective. This enables direct measurement of the system's defect detection capability independent of production defect prevalence.

The following metrics require knowledge of baseline defect rate and are measured during production operation on mixed populations of good and defective lenses:

- **Accuracy:** $(True\ Accepts + True\ Rejects) / Total\ Samples$
- **Precision:** $True\ Rejects / (True\ Rejects + False\ Rejects)$

- **Recall (Equivalent to TRR):** True Rejects / (True Rejects + False Accepts)
- **F1-Score:** $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

These performance metrics will be applied to the overall pass/fail determination and classification of each artifact. The True Accept Rate will be used to evaluate True Accept conditions, stratified by artifact. For example, there is a different True Accept Rate for pristine images, acceptable-sized air bubbles, and acceptable small debris. The True Reject Rate will be used to assess true reject conditions, stratified by debris type. For example, there is a different True Reject Rate for HEMA, fiber, extraneous polymer, and FM.

Section 6: Algorithm Requirements

It is recommended that two algorithms be developed to reduce the requirements for each algorithm solution. If only one algorithm is developed, it must meet the requirements of both the Edge Algorithm and the Back Office Algorithm. The Edge Algorithm will focus on delivering speed and accuracy of overall pass/fail determination, while the Back Office Algorithm will focus on classifying and measuring individual debris occurrences. The scope of this document only pertains to the edge algorithm, for more information on the back office algorithm reference the Back Office Algorithm URS (**Doc # TBD**).

6.1 Edge Algorithm

The Edge Algorithm shall complete the Pass/Fail determination and the probability of pristine bowl presentation within 200 milliseconds. The Edge Algorithm shall output a Pass/Fail determination of an input image that meets the following criteria:

$$\text{True Acceptance Rate (TAR)} = 99.9\%$$

$$\text{True Reject Rate (TRR)} = 99.7\%$$

6.2 Acceptance Criteria and Sample Size for Severity Level 4 Attributes

Per VWA-0523, all FM occurrences (in solution and on the lens) are severity level 4 attributes and must meet the claim of being 99% confident that the probability of a primary package not having FM greater than 647 microns is at least 95%. Chart from VAL-0009(14.0).

Column 1	Column 2	Column 1	Column 2	Column 1	Column 2	Column 1	Column 2
Lenses Inspected	Max. No. of Failures	Lenses Inspected	Max. No. of Failures	Lenses Inspected	Max. No. of Failures	Lenses Inspected	Max. No. of Failures
90 to 129	0	425 to 450	11	681 to 705	21	925 to 948	31
130 to 164	1	451 to 477	12	706 to 729	22	949 to 972	32
165 to 197	2	478 to 503	13	730 to 754	23	973 to 996	33
198 to 228	3	504 to 528	14	755 to 778	24	997 to 1019	34
229 to 258	4	529 to 554	15	779 to 803	25	1020 to 1043	35
259 to 287	5	555 to 579	16	804 to 827	26	1044 to 1067	36
288 to 315	6	580 to 605	17	828 to 851	27	1068 to 1090	37
316 to 343	7	606 to 630	18	852 to 876	28	1091 to 1114	38
344 to 370	8	631 to 655	19	877 to 900	29	1115 to 1138	39
371 to 397	9	656 to 680	20	901 to 924	30	1139 to 1161	40
398 to 424	10						

Figure 18. Table from VAL-0009(14.0) showing the Sample Sizes and Acceptance Criteria for 95% Probability of Success with 99% Confidence.

Section 7: Functional & Process Requirements

Attribute	Requirement Description
SKU range	FMD model must be sufficiently trained to meet performance criteria on the entire range of SKUs within a product line
Intensity range	FMD model must be sufficiently trained to meet performance criteria across the backlight variation of the 3GT manufacturing lines.
Focus and aperture	Camera and lens are pre-set to have the correct focus and aperture
Trigger sensor supervision	The Vision System must provide a function to monitor the trigger sensor functionality
Save images	System must be configurable to save images
Pass Fail output	The FMD system must output pass/reject signal for each individual package from an array
Ethernet Comms.	The system must utilize ethernet communication to connect to PLC
Documentation	Communication architecture must be documented in the SDS

Attribute	Requirement Description
Spare Computer	Spare computers must be pre-imaged with all required software.
Fail safe – Result buffer	The result buffer must be configured fail safe. In event of a failure HMI will alarm.
Fail safe - Protocol	Communication protocol must be designed for fail safe communications.
Pallet configuration	System must be configurable to inspect 10, 12 or 15 packages for every index under the cameras
Camera Focus	Cameras are set up using an objective measure of focus.
Camera Focal depth	Camera focal depth must include the entire of the primary package bowl
Timeout	System returns a “reject” status if the image analysis exceeds an expected time and array will be rejected
Live mode	User interface must contain a live view mode
Statistics	User interface displays FM pass/fail statistics of the historical data per shift
System Layout	UI must contain a home screen that includes statistics, system status indicator, and live view of results
Image capture	UI includes a mode to capture an adjustable number of images to a directory on the computer
Offline Capability	The system shall have the capability to run offline to review images or for algorithm assessment
Configuration parameters	The UI must have configurable parameters for the ASComm CIP for: - IP address of PLC - CPU slot position on rack - Update rate of data polling in range of 10 – 100ms - Transaction Timeout [ms] - Connection Timeout [ms]
Configuration access	The configuration of the UI must only be accessible for Administrator users
System ready	HMI or user interface shall display the ready/not-ready signal

Attribute	Requirement Description
Conveyor index	The FMD system must be in ready state to index the saline buffer.
CarrierID	CarrierID must be generated in the PLC and transferred to the FMD system to receive the pass-fail results along with the CarrierID to apply to the appropriate tracking location.
Pass / Fail results	Pass fail results for individual package along with its carrier ID must be displayed on the software tracking screen on the HMI
Tagging product	Software shall be able to independently tag each array for accept/reject
Consecutive failure	Zone must be faulted out with an alarm when FMD system fails 3 consecutive array failure is detected.
Heat-seal rejects	FMD system rejects must be heat-sealed upon confirmation of correct foil placement and rejected into the reject chute.
Algorithm Name	Algorithm Name must be hardcoded in the PLC on the configuration data set to allow for transfer to FMD system and also displayed on the HMI.
Algorithm Checksum	Algorithm Checksum must be hardcoded in the PLC on the configuration data set to allow for transfer to FMD system
Framework Checksum	Framework Checksum must be hardcoded in the PLC on the configuration data set to allow for transfer to FMD system
Statistics	Reject counts for FMD must be calculated and displayed on the HMI for current and previous shift
Cycle time	Cycle time for FMD station must be calculated and displayed on the HMI
Yield	FMD yield must be calculated and displayed on the HMI for current and previous shift
Network	Network architecture must be updated with connection status and displayed on the HMI
Network connection	Alarm must be configured when network is disconnected and displayed on the HMI

Section 8: Data Requirements

There are no data specific requirements associated with this URS. The system is not a formal data repository.

Section 9: Reporting Requirements

This section is not applicable. FMDVS does not have reporting capability.

Section 10: Compliance Requirements

There are no compliance specific requirements associated with this URS. Not applicable; the system does not impact compliance aspects of the contact lens/medical device. 3GT FM Detection Vision System represents an additional inspection step that does not replace any existing controls or specifications pertinent to the registration of the device.

Section 11: Technical Requirements

11.1 Edge Algorithm Hardware Requirements

The Edge Algorithm will reside on hardware identified by the algorithm vendor, which will interface directly with the image formation hardware, programmable logic controllers in a manufacturing environment, and Operational Technology (OT) systems. The algorithm hardware selection is in scope, but the interactions between the algorithm hardware and other systems are out of scope.

The Edge Algorithm hardware solution *shall*:

- be capable of operating 24 hours a day, 7 days a week, with minimal interruptions for scheduled maintenance.
- have a mean time between failures (MTBF) estimated to be 30,000 hours.
- be compatible with U.S. and European power systems, 120 VAC 60 Hz and 230 VAC 50 Hz.
- *shall* be capable of interfacing with up to four machine vision standard 10GigE devices.

11.2 Operating Capabilities and Reliability Requirements

Development of deployment software and operator interfaces is out of scope for this URS. However, Edge Algorithm outputs shall be sufficient to support production integration,

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monitoring, diagnostics, fault handling, and data retention requirements defined by the applicable deployment-software or interface specification (DOC ID). The software will:

- be able to automatically recover from sudden interruptions and the faults should be logged in a report.
- provide configurable warnings and alarms.
- display and store all identified defect codes such as holes edge tears foreign matter ect.
- display the trends of the lens defects as frequency of occurrence over a time period.
- provide system-related diagnostic information, CPU usage, memory usage, processing time, data traffic, and I/O status.